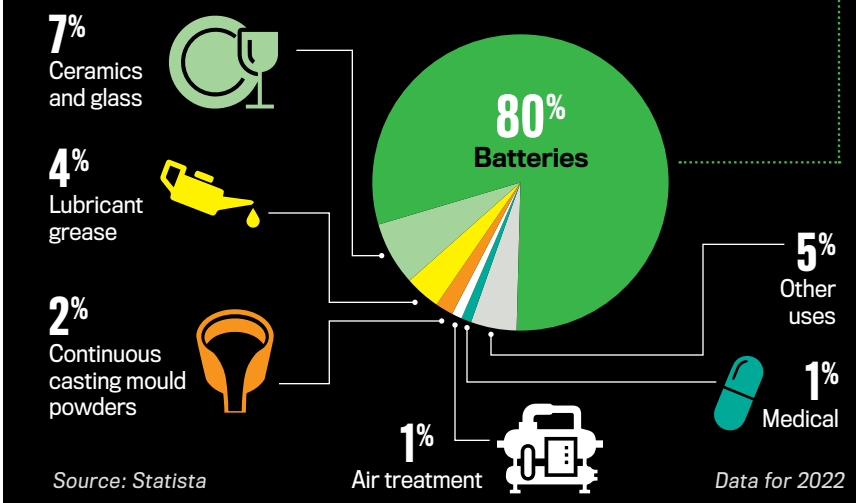
The discovery is significant in the face of the Modi government's production-linked incentive (PLI) scheme for manufacturing advanced chemistry cell (ACC) battery storage. Four companies—Rajesh Exports, Reliance New Energy Solar, Hyundai Global Motors Company and Ola Electric Mobility—have been selected for incentives worth $2.4 billion (Rs 18,100 crore).

India needs investments worth up to $4.5 billion (Rs 33,750 crore) to achieve the PLI target of setting up 50 gigawatt hours (GWh) of lithium-ion cell and battery manufacturing plants, according to a study titled 'How can India indigenise lithium-ion battery manufacturing?' by CEEW. The country requires up to 903 GWh of energy storage to decarbonise its mobility and power sectors by 2030 and Li-ion batteries will meet the majority of this demand. "However, it is important to develop mineral processing and raw material processing capability to truly become *aatmanirbhar*," adds Jain.

AREAS OF CONSUMPTION

Distribution of lithium end-usage worldwide



EVs DRIVE THE HUNT

- India requires up to 903 GWh of energy storage to decarbonise its mobility and power sectors by 2030. Li-ion batteries will meet the majority of this demand
- Electric vehicles (EVs) saw a rise in sales, with 223 per cent growth in 2022, accounting for 48,000 units
- The EV expansion aims to reach 40 per cent of two-wheelers and private cars, and almost 100 per cent of commercial buses by 2030
- Last year, the Centre announced a

The caution reflects the many technical and physical challenges involved, including costly exploration processes and the lack of experience in lithium extraction. Others point to administrative inertia and the fact that it took the GSI two decades to reach the prospecting stage (G3) and get an inferred estimate. The GSI had come out with its initial survey after the reconnaissance stage (G4) in 1999, but no further work was carried out until two years ago when its team descended on Salal village. "Following the National Mineral Exploration Policy, 2016, the GSI has given thrust on exploration of strategic and critical minerals in addition to bulk minerals," says Dr S. Raju, director general, GSI. The survey concluded that 'higher values of lithium are persistent throughout the belt in the bauxite column'. The 69-page 'Final Report on Regional Geochemical Survey for Base Metals and Lithium in Salal Area' said 'lithium prospect in the bauxite column in the area of investigation appears to be promising' and 'need [s] to be taken for further detailed work'.

The exploratory search goes through four stages—G4, G3, G2 and G1—before the final extraction. Now, the Salal site will undergo G2 (general exploration), which entails closed-space drilling within a radius of 100-200 metres with vertical and lateral extension to determine the accurate extent of ore reserves and consequential grading for mining. In February 2021, a similar 'inferred' category with an estimate of 1,600 tonnes was found in the Marlagalla-Allapatna area of Karnataka's Mandya district. And in February this year, Union minister for coal and mines Pralhad Joshi told the Lok Sabha that the Atomic Minerals Directorate for Exploration and Research, a constituent unit of the Department of Atomic Energy, is carrying out exploration for lithium in parts of Mandya and Yadgir districts in Karnataka. But the resource has not yet been augmented by the GSI. "Indifference, delay and budgetary constraints have held us back from doing systematic survey and mapping," says a retired senior GSI official.

Prof. Pankaj Srivastava, who teaches mineral exploration and economic geology at Jammu University, says the G2 stage can give 'indicated' results of grade, quality and quantity of the mineral with a chance of error reduced to 20-30 per cent. In the current inferred estimate, the margin of error could be as high as 50 per cent. Even if the exploration process is expedited, it will take a minimum of 5-6 years to extract minerals. "There is a low level of confidence at G3. Now the Union Territory government has to decide about the further course of action (at G2 and G1 stages) as the GSI's role is over. It requires large-scale investment and they can get it done through public sector enterprises or private entities depending on them," adds Srivastava.

The UT government has assigned a team of experts to study the